Programming Fundamentals (PF) – Section C



Assignment Title: Function Task
Subject Name: Programming Fundamentals Lab
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In-Lab Tasks: Function Practice

Task 1 (ATM Management System Using Functions): Write a C program that simulates a simple ATM system using five user-defined functions. Create `welcome()`, which takes no arguments and returns nothing, to display a welcome message. Create `displayAmount()`, which accepts a withdrawal amount, returns nothing, and displays the amount. Create `getWithdrawalLimit()`, which takes no arguments and returns the predefined maximum withdrawal limit of Rs. 50,000. Create `calculateBalance()`, which accepts the current account balance and the withdrawal amount and returns the remaining balance. Create `checkWithdrawal()`, which accepts the current balance, withdrawal amount, and maximum limit, and returns 1 if the withdrawal is valid or 0 otherwise. A withdrawal is valid only if the amount is greater than zero, does not exceed the account balance, and does not exceed the maximum withdrawal limit. In `main()`, call the functions as required, display the relevant information, and update the balance only if the withdrawal is valid.

main.c

Run

```c
#include <stdio.h>
void welcome(){
    printf("Welcome to ATM Management System!\n");
}
void displayAmount(float amount){
    printf("Withdrawal Amount: Rs. %.2f\n", amount);
}
float getWithdrawalLimit(){
    return 50000;
}
float calculateBalance(float balance, float amount){
    return balance - amount;
}
int checkWithdrawal(float balance, float amount, float limit){
    if (amount > 0 && amount <= balance && amount <= limit){
        return 1;
    }
    else{
        return 0;
    }
}
int main(){
    float amount, limit, balance = 80000;
    int valid;

    welcome();
    printf("Current Balance: Rs. %.2f\n", balance);

    limit = getWithdrawalLimit();

    printf("Maximum Withdrawal Limit: Rs. %.2f\n", limit);
    printf("Enter withdrawal amount: Rs. ");
    scanf("%f", &amount);
```

```c
    printf("Maximum Withdrawal Limit: Rs. %.2f\n", limit);
    printf("Enter withdrawal amount: Rs. ");
    scanf("%f", &amount);

    valid = checkWithdrawal(balance, amount, limit);

    if (valid == 1) {
        displayAmount(amount);

        balance = calculateBalance(balance, amount);

        printf("Withdrawal Successful!\n");
        printf("Remaining Balance: Rs. %.2f\n", balance);
    }
    else {
        printf("Invalid Withdrawal!\n");
        printf("Please check the amount and try again.\n");
        printf("Your Balance is still Rs. %.2f\n", balance);
    }
    return 0;
}
```

Output

```
Welcome to ATM Management System!
Current Balance: Rs. 80000.00
Maximum Withdrawal Limit: Rs. 50000.00
Enter withdrawal amount: Rs. 5000
Withdrawal Amount: Rs. 5000.00
Withdrawal Successful!
Remaining Balance: Rs. 75000.00
```

Task 2 (Shopping Management System Using Functions): Write a C program that simulates a shopping management system using user-defined functions. The program should display a welcome message, show a

menu of available products with their prices, allow the user to select products and enter quantities, calculate the total price of each product, calculate the subtotal of all selected products, apply a discount if the subtotal reaches a specified amount, calculate the final bill, and display a shopping receipt. Create separate functions to display the welcome message, show the product menu, get the product price, calculate the price of an item, calculate the subtotal, calculate the discount, calculate the final bill, display the receipt, and ask the user whether they want to continue shopping. Use functions with different combinations of arguments and return types, including functions that take no arguments and return no value, functions that take arguments and return no value, functions that take no arguments and return a value, and functions that take arguments and return a value. Use appropriate loops and conditional statements to manage product selection, quantities, discounts, and repeated shopping. Display the selected products, quantities, individual prices, subtotal, discount, and final payable amount.

main.c Run ▶

```c
#include <stdio.h>
void welcome(){
    printf("Welcome to Shopping System!\n");
}
void menu(){
    printf("\n1. Burger - Rs. 450\n");
    printf("2. Pizza - Rs. 1200\n");
    printf("3. Fries - Rs. 250\n");
}
int getPrice(int choice){
    if (choice == 1)
        return 450;
    else if (choice == 2)
        return 1200;
    else if (choice == 3)
        return 250;
    else
        return 0;
}
int itemTotal(int price, int quantity){
    return price * quantity;
}
int discount(int subtotal){
    if (subtotal >= 2000)
        return subtotal * 10 / 100;
    else
        return 0;
}
int main(){
```

main.c

Run

```c
int main(){
    int choice, quantity, price, total;
    int subtotal = 0, dis, bill, again = 1;

    welcome();
    while (again == 1)  {
        menu();

        printf("Enter choice: ");
        scanf("%d", &choice);

        price = getPrice(choice);

        if (price == 0) {
            printf("Invalid choice!\n");
            continue;
        }
        printf("Enter quantity: ");
        scanf("%d", &quantity);

        if (quantity <= 0)    {
            printf("Invalid quantity!\n");
            continue;
        }
        total = itemTotal(price, quantity);
        subtotal = subtotal + total;
        printf("Item Total: Rs. %d\n", total);
        printf("Continue shopping? (1=Yes, 0=No): ");
        scanf("%d", &again);
```

```
    }
    dis = discount(subtotal);
    bill = subtotal - dis;

    printf("\n----- RECEIPT -----\n");
    printf("Subtotal: Rs. %d\n", subtotal);
    printf("Discount: Rs. %d\n", dis);
    printf("Final Bill: Rs. %d\n", bill);

    return 0;
}
```

Output

```
Welcome to Shopping System!

1. Burger - Rs. 450
2. Pizza - Rs. 1200
3. Fries - Rs. 250
Enter choice: 1
Enter quantity: 4
Item Total: Rs. 1800
Continue shopping? (1=Yes, 0=No): 1

1. Burger - Rs. 450
2. Pizza - Rs. 1200
3. Fries - Rs. 250
Enter choice: 3
Enter quantity: 2
Item Total: Rs. 500
Continue shopping? (1=Yes, 0=No): 0

----- RECEIPT -----
Subtotal: Rs. 2300
Discount: Rs. 230
Final Bill: Rs. 2070
```